Tribhuvan University

Faculty of Humanities and Social Sciences

ONLINE DOCTOR APPOINTMENT SYSTEM

Submitted To

Department of Computer Application

Trinity International College

Submitted By

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DECLARATION

A project entitled “Online Doctor Appointment System” is being submitted to the Department of Computer Application, Trinity International College, Dillibazar, Kathmandu, Nepal for the fulfillment of the Fourth semester under the supervision of Mr. Dharma Raj Poudel. This project is original and has not been submitted earlier in part or full in this or any other form to any university or institute, here or elsewhere, for the award of any degree.

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RECOMMENDATION

This is to recommend that **Ronika Humagain** and **Simran Pun** have carried out research entitled **"Online Doctor Appointment System"** for the fulfillment of the Fourth semester in Bachelor's degree of Bachelor's in Computer Application under Tribhuvan University under my supervision. To my knowledge, this work has not been submitted for any other degree.

They have fulfilled all the requirements laid down by the Department of Humanities and Social Sciences Trinity International College, Dillibazar, Kathmandu, Nepal.

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Faculty of Humanities and Social Sciences

Trinity International College

LETTER OF APPROVAL

This is to certify that this project prepared by **Ronika Humagain** and **Simran Pun** entitled **"ONLINE DOCTOR APPOINTMENT SYSTEM"** in partial fulfillment of the requirement for the degree of Bachelor in Computer Application has been evaluated. In our opinion it is a satisfactory in scope and quality as a project for the required degree.

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Internal Examiner	External Examiner

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We also wish to express our sincere thanks to Trinity International College for providing and giving us permission to use all required equipment and the necessary material to complete the project.

Our friends and family deserve a special mention for their encouragement, love, and support. Their unwavering support helped us stay motivated throughout the project.

- -

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Simran Pun, 26164/2022

ABSTRACT

The goal of Online Doctor Appointment System is to mechanize the prevailing manual system by the assistance of computerized equipment's and developed computer application, fulfilling their needs, in order that their valuable data/information may be hold on for an extended period with simple retrieving and handling of the identical. The mandatory software and hardware are effortlessly obtainable and easy to work with them. This will involve an iterative process to make the system usable and easy to use by the user.

Online Doctor Appointment System, as defined higher than, will cause error free, protected, consistent and quick management system. It will assist the user to target their alternative activities rather to target the record keeping. Therefore, it will facilitate organization in higher deployment of resources. The organization will maintain computerized records while not redundant entries which means that one need not to be distracted by data that is not relevant, whereas having the ability to achieve the knowledge.

Keyword: Online Doctor Appointment System

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List of Abbreviation

CSS - Cascading Style Sheets

FCFS - First Come First Serve

GUI - Graphical User Interface

HTML - Hypertext Markup Language

MySQL - My Structured Query Language

PHP- Hypertext Preprocessor

RDBMS - Relational Database Management System

SQL - Structured Query Language

SRS - System Requirement Specification

TC - Test Case

UI - User Interface

WAMP - Windows, Apache, MySQL, and PHP

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Chapter 1

INTRODUCTION

1.1 Introduction

In recent years, information systems have become critical components of any expanding firm. Because a growing business need accurate information and the essential technology to identify challenges and keep up with ever-increasing client demands, Data System Technology has been a crucial force for organisations to see their business requirements. Businesses increasingly use data systems and available technology because they understand the value of electronically storing and changing information.

The Online Doctor Appointment System is created to address the challenges that existed within the active manual system. [1] This system is intended to eliminate or, in some situations, reduce the difficulties encountered by the previous method. It also delivers error warnings when encountering invalid data. Formal expertise is not necessary for the user to operate this system. As a result, all of this demonstrates that it is user friendly. This system will enable the centre acquire, update, and report patient information more efficiently, allowing clinicians to make timely and successful diagnoses. At the same time, the centre can use this technology to track their patients' medical histories. Improving patient care management is one of the primary goals of the healthcare business today in order to improve the global healthcare system. As the population increases, so does the demand for healthcare services and options. Globally, health care service providers are under increasing pressure to cut costs while also improving accessibility and quality of care, particularly in resolving long waiting times, delays, and patient queues. As a result, a Patient Scheduling System is introduced as a key component for scheduling and managing appointments. Specifically, online scheduling software has streamlined and automated the healthcare administration process for organisations of all sizes. Each organisation, large or small, faces issues in preserving information about appointments, doctor bookings, doctor fees, and doctor schedules. Using an information system to manage information in health care, such as patient records, patient appointment systems, patient scheduling appointments, and doctor schedules, is not only a simpler way to save time and money, but it is also a way to support and improve health care information so that it is more accessible and flexible (modifying, saving, deleting, updating) for system users while also storing data efficiently. It also increases the quality of data control.

1.2 Problem Statement

The traditional healthcare appointment system is often plagued by inefficiencies, long waiting times, and limited accessibility, hindering patients' ability to promptly and conveniently access healthcare services. In the contemporary digital era, the need for an advanced and user-friendly Online Doctor Appointment System is paramount to address these challenges and enhance the overall healthcare experience.

Physical appointment scheduling processes pose challenges for patients, especially those with mobility issues, busy schedules, or residing in remote areas. Lack of an online system limits the accessibility of healthcare services, leading to delays in seeking medical attention. The conventional method of scheduling appointments over the phone or in person is time-consuming and often results in long wait times for patients. Manual scheduling may lead to errors, confusion, and difficulties in managing appointment records efficiently. The absence of a digital platform hinders effective communication between patients and healthcare providers. Patients may miss out on important updates, reminders, and health-related information due to the lack of an integrated online system.

The project aims to address these challenges by developing a secure and efficient Online Doctor Appointment System. This system will leverage modern technologies, including secure data encryption, user-friendly interfaces, and real-time communication tools, to provide patients with a seamless and convenient way to schedule appointments.

1.3 Objective

- To manage the information of Doctors and Patients and let the patients' book online appointment.
- To improve accessibility to healthcare services.
- To increase overall patient satisfaction.

1.4 Scope and Limitations

1.4.1 Scope

Growth in patients is undoubtedly good for the healthcare industry, but it also presents difficulties for personnel and facility administrators. Methods that worked well in the past might not be able to handle the increase of patients. The following is the scope of the Online Doctor Appointment System:

- The patient scheduling system replaces traditional booking chaos with online facility.
- The online scheduler which contains all the doctors' available time of appointment.
- Information update, review, and maintenance.
- For patients, only information related to their reservation will be displayed through their account website.
- For clinic staffs, the information of both doctors and patients can be reviewed on system's main server.
- The communication between the staff is done using the local clinic network.
- Have a good user interface.
- Easy to understand by the user and operator.
- To utilize resources in an efficient manner by increasing their productivity through automation.

1.4.2 Limitations

Even though we have made every attempt to make the program flexible and simple to use, there are some limitations that we are unable to overcome. Although great efforts have been taken to make the software simple to use even for those who are not familiar with computers, it is acknowledged that an average person may initially find it a little challenging. Help is provided to the user at every stage to make using the software as convenient as possible. The following are the only limitations of the Online Doctor Appointment System:

- After the patient's appointment, billing statements and receipts are generated manually.
- Additional details unrelated to scheduling an appointment are not accessible.

1.5 Development Methodology

The Iterative Waterfall Model is a variation of the traditional Waterfall Model used in software development, where the process is still divided into distinct phases—such as requirements, design, implementation, testing, and maintenance—but each phase can be revisited and repeated based on feedback or errors discovered in later stages.

This methodology used in the Online Doctor Appointment System is Iterative Waterfall Model methodology.

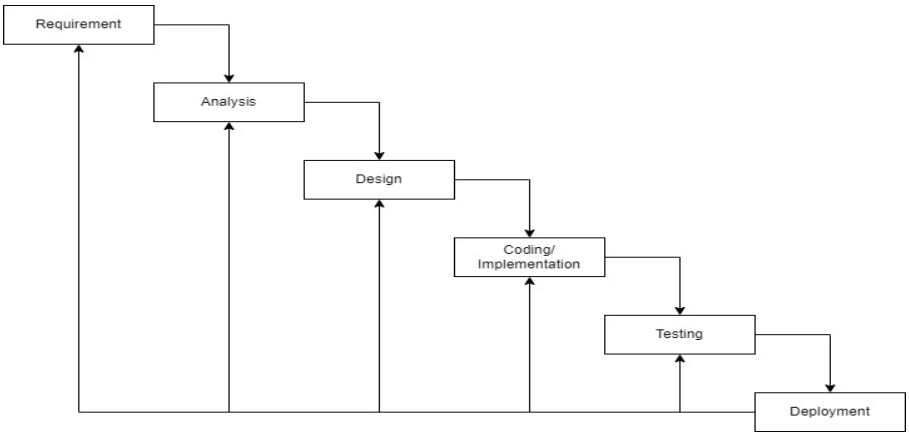


Figure: 1.5 Iterative waterfall model

1.6 Report Organization

In an online doctor appointment system**,** organizing reports is essential for monitoring activities, improving healthcare service quality, and enabling informed decision-making. Reports should be structured in a logical and user-friendly manner, tailored to the needs of different stakeholders such as patients, doctors, and admin**.**

Chapter 1:

This chapter describes the introduction of Online Doctor Appointment System. It also explains the objectives, scope and limitations, Methodology.

Chapter 2:

This chapter consists of a background study where we look at the research done in this field before and talk about similar systems that already exist for online doctor appointment system.

Chapter 3:

This chapter consists of a system analysis and system design.

Chapter 4:

This chapter contains implementation and testing process.

Chapter 5:

This chapter concludes the report and provides insights into the project, discusses lesson learnt and the future recommendation of the projects.

Chapter 2

Background Study and Literature Review

2.1 Background Study

Scheduling appointments has grown more difficult, particularly for medical personnel working in clinics and hospitals. These issues could arise for a variety of reasons, such as a physician who practices in multiple clinics and frequently travels between medical facilities or an excessive volume of patient traffic. In addition, overlapping appointments, an increase in no-shows, general patient unhappiness, and revenue loss for healthcare facilities can result from inefficient appointment administration.

Nowadays, a lot of healthcare facilities combine computerized appointment scheduling with phone-based scheduling. Although this combination with outsourcing services results in a more effective system, there are still some opportunities for human and technical error. This online tool is a useful addition to the website of any hospital or clinic. It lessens the effort involved in running a medical center. Other benefits include having more time to devote to medical care, improved patient compliance, and financial viability. Representing patient satisfaction and revenue gains is the primary goal of an effective online patient scheduling system. A bridge connecting timely access to effective healthcare services and an active appointment scheduling system. The goal of the suggested patient scheduling system is to streamline processes and lessen crowding in waiting areas.

Any healthcare facility that manages patient ratings and obligations runs the danger of overspending on patient appointments. Especially for tiny clinics with doctors running their own practices, it makes no sense to stick with the antiquated paper-based scheduling system. The more handwritten notes and cancellations there are in appointment books, the more this procedure confuses and irritates the staff members in charge of keeping track of them because they are time-consuming and limited. Web-based patient scheduling is quicker and supports numerous users at once when compared to paper-based appointment scheduling. Web-based appointment scheduling reduces the number of no-shows by generating appointment records and email reminders.

2.2 Literature Review

Overview of Similar Project

- **HSC Medical Center Appointment System:**

Nowadays the need to queue up for a long time to consult a medical officer has minimized by technological facilities available over the Internet. HSC Medical Center is one such organization that has adapted to the current technology era. The HSC Medical Center's appointment system does not require an ID or password to log in before scheduling an appointment, and appointments are only good for 24 hours. To complete the appointment, users must fill out a form and click the submit button. [2]

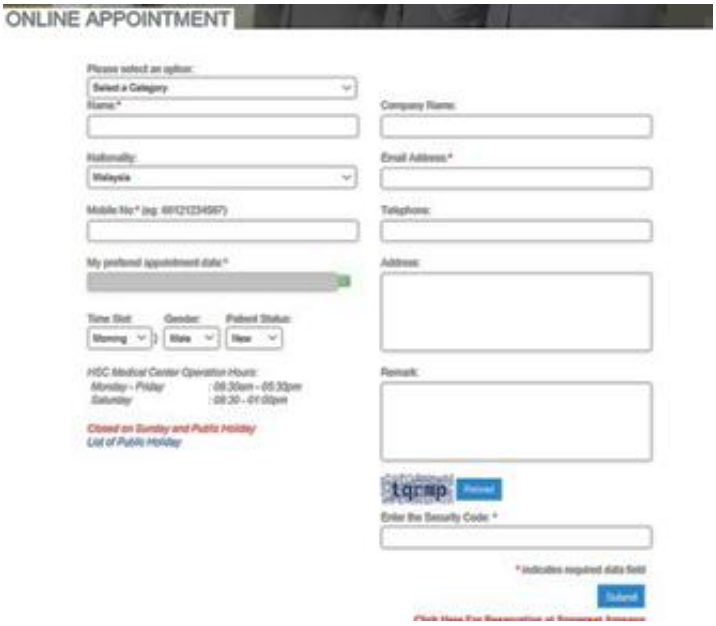
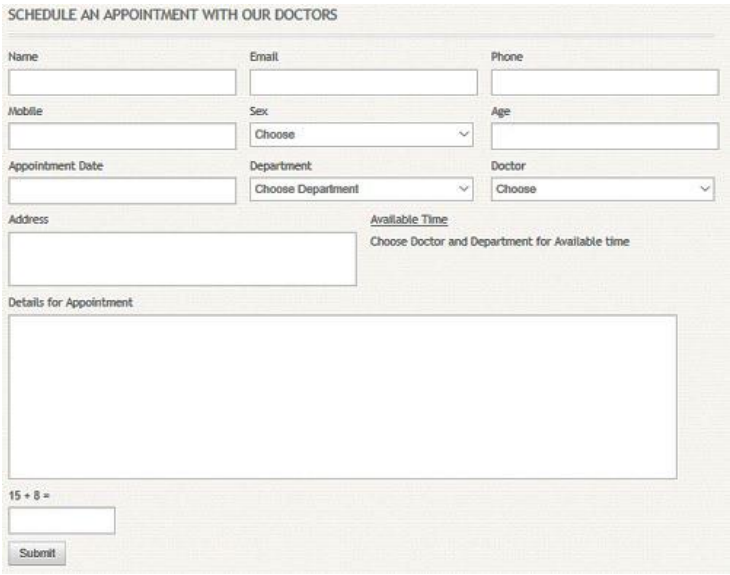


Fig 2.2 (a): HSC Medical Centre Appointment System

- **OM Hospital Appointment system:**

Om Hospital has an online system for scheduling appointments. It was created in an attempt to lower the increased cost by employing the inefficient phone. The user can enter the appointment time in this system at any moment. The user merely needs to enter their information and submit the form; no ID or password is needed to utilize this system. The availability of the doctor and department can also be checked by the patient. [3]



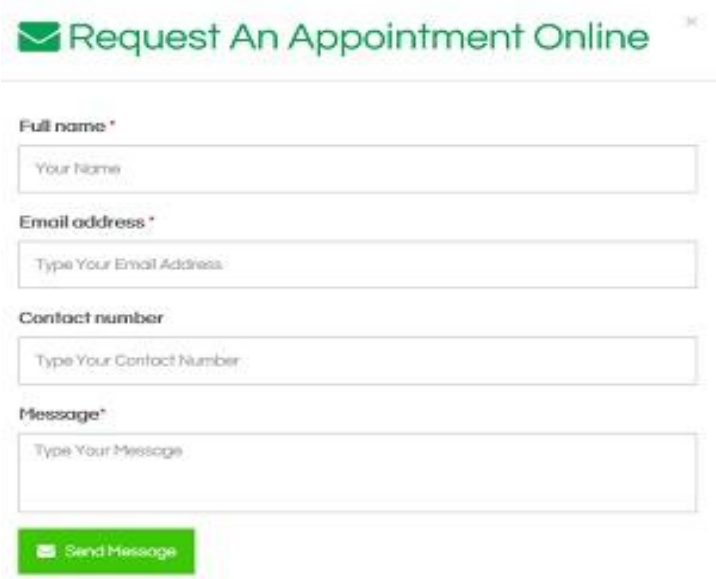


Fig 2.2 (b): Om Hospital Appointment System[https://www.omhospitalnepal.com/]

Chapter 3

System Analysis and Design

3.1 System Analysis

Various needs are taken into consideration before the new system's development gets underway. Since our project is only focused on healthcare, the requirement analysis is completed with an emphasis on the many healthcare facets that are common in the nation. An essential step in the creation of any system is requirement analysis or study. A thorough analysis of the procedure needs to be done using a variety of methods, such as questionnaires and interviews. To draw a conclusion, the information gathered from various sources needs to be examined. An awareness of the system's operation is the conclusion. We refer to this system as the current system. Currently, the current system is being closely examined, and issues are being found. Initial research is a problem-solving process that necessitates close collaboration between system developers and consumers. It does different kinds of feasibility studies. From these investigations, an approximate picture of the system's operations may be gathered, from which tactics for an efficient system research and analysis can be chosen. The mechanism is examined in great detail and examined. The following outline is used to complete the requirement analysis:

3.1.1 Requirement Analysis

i. Functional Requirements:

- Users profile management and registration.
- Patients to make online appointment, look their previous health records and doctor's prescriptions.
- Doctors to give appointments, e-prescriptions, and view patient's history.
- Site should have details online help manual for patients. Local language support is essential.
- Admin/staff to take backup of all kind of data, view log and generate system reports.

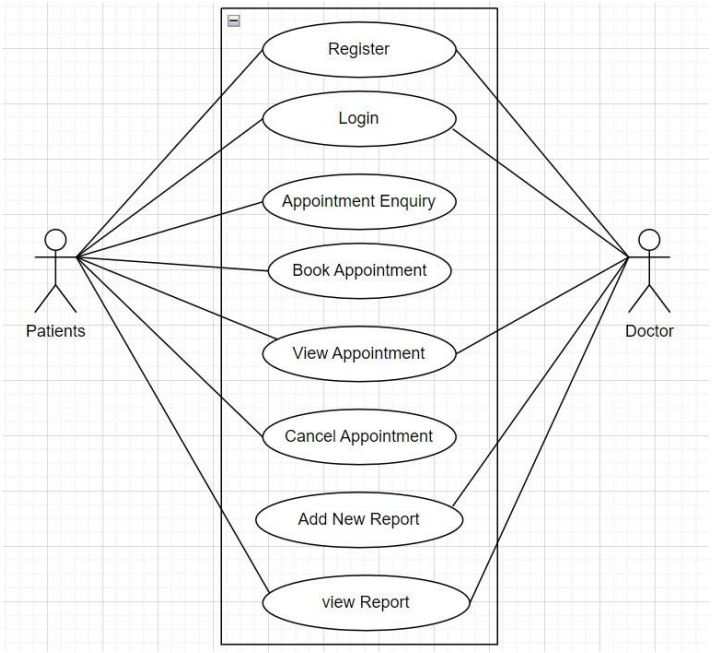


Fig 3.1.1 (a): Use Case Diagram of Doctor and Patient

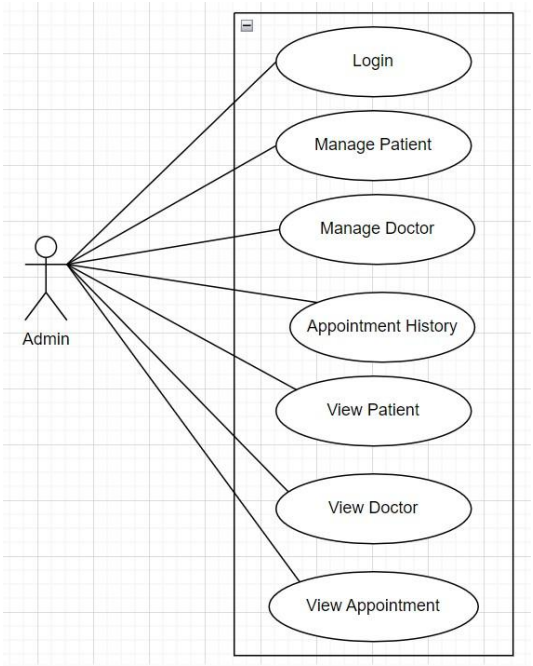


Fig 3.1.1 (b): Use Case Diagram of Admin

ii. Non-Functional Requirements:

- Secure access of confidential data (user's details).
- 24 X 7 availability.
- Better component design to get better performance at peak time.
- Flexible service-based architecture will be highly desirable for future extension.

3.1.2 Feasibility Study

A feasibility analysis evaluates how feasible a project is as is. It gives the suggested project's level of viability. A feasibility analysis assists us in assessing the project's worth, identifying a market for the project, determining its financial viability, and ultimately determining whether or not it is worthwhile to devote time and resources to the project. A feasibility analysis, to put it briefly, assesses the project's likelihood of success. Therefore, a system's ability to function, impact on the organization, ability to meet user requirements, and improved resource usage are all tested via feasibility analysis. The feasibility analysis that follows was completed before beginning the project:

i. Technical Feasibility

This includes researching performance, function, and potential roadblocks to the creation of a workable system. We examined all of the functionality that was specified in the System Requirement Specification (SRS) for this feasibility study, and we used various frontend and backend platforms to see if everything could be implemented.

The entire web-based program known as Project Online Doctor Appointment System is available. The Online Doctor Appointment System is primarily connected to the following technologies and tools:

- HTML
- CSS
- PHP
- MySQL

The technical skills needed to use each of these technologies are achievable and they are all publicly available. The ease of implementation with these technologies and the time constraints for system development are in harmony. This demonstrates the technical viability of the Online Doctor Appointment System project.

ii. Operational Feasibility

There is no doubt that the proposed system is entirely GUI-based, extremely user-friendly, and all inputs are self-explanatory, even for the average person. If and when an issue emerges, the target users can search for assistance using the online application. Operational feasibility is the ratio of a system's ability to meet requirements, capitalize on opportunities found during the requirement analysis phase, and resolve problems to the degree that the system is operationally feasible. This program resolves the client's issues, which are listed in the problem statement above.

iii. Economic Feasibility

This is a crucial step that must be completed when creating a project. It has to do with whether the suggested enterprise can actually produce financial gains. There is a benefit-cost analysis. We selected the technology with the lowest feasible cost in mind. An inquiry into an undertaking's costs and earnings with the aim of determining if it is cohesive and imaginable to finish is known as economic feasibility or financial possibility. This application is financially feasible because it was created with Dreamweaver and other resources that are freely available for download from the Internet.

iv. Schedule Feasibility

The system's design prioritizes simplicity and ease of use, allowing for efficient development. The project is completed within a sixth week.

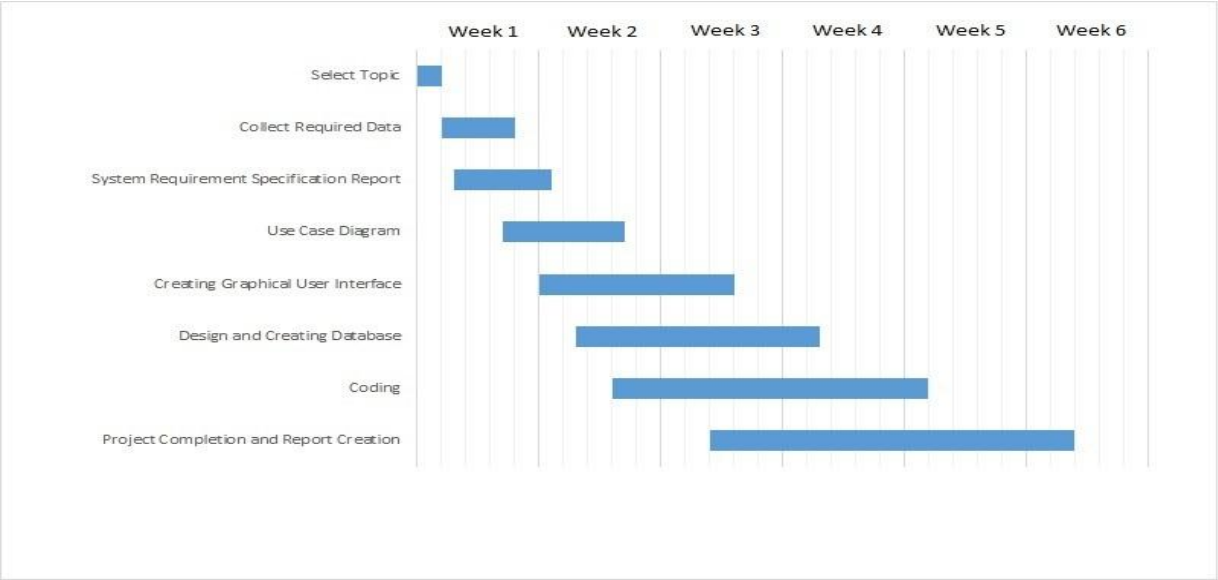


Figure: 3.1.2 Gantt Chart

3.1.3 Data Modelling: ER Diagram

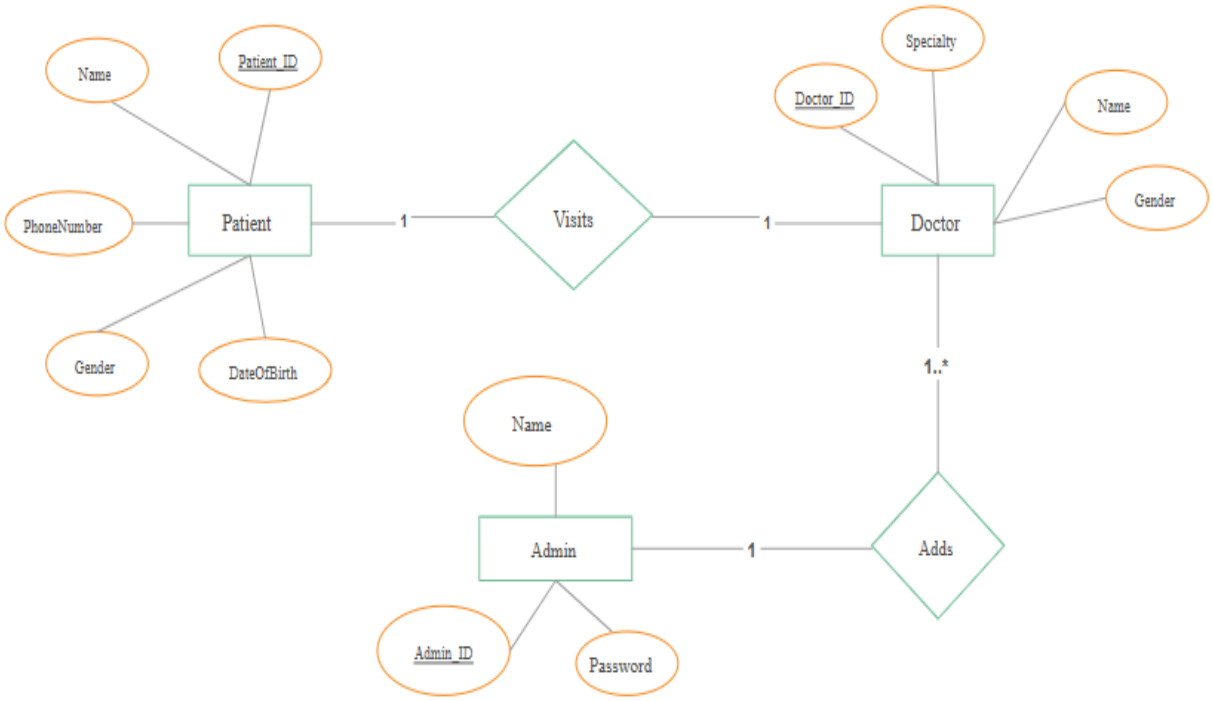


Figure: 3.1.3 ER diagram of Online Doctor Appointment System

3.1.4 Flowchart

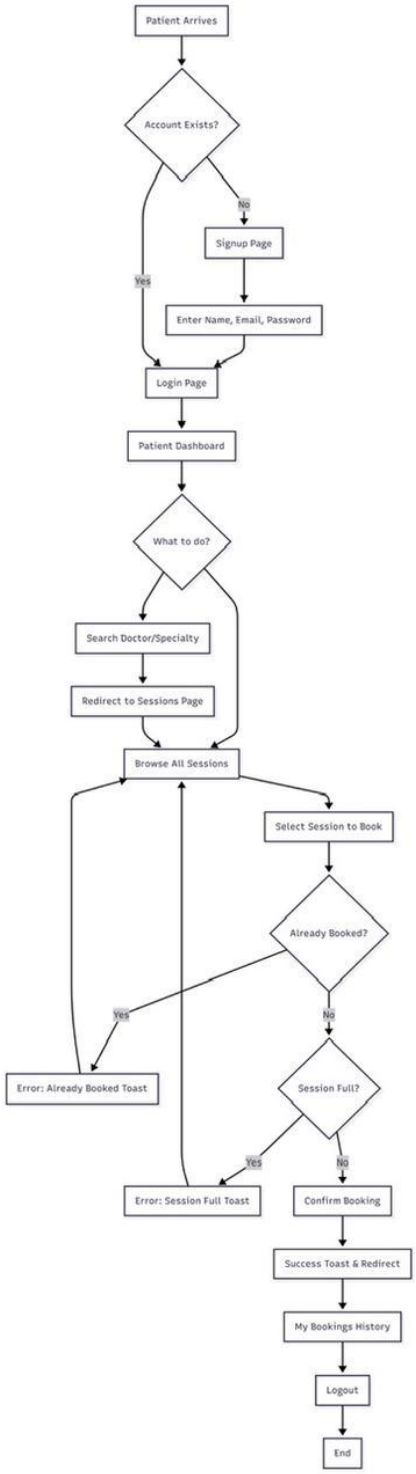


Figure: 3.1.4 Flowchart

3.2 System Design

System design refers to the process of defining the architecture, components, modules, interfaces, and data for a system to satisfy specified requirements. It involves designing the overall structure of a software or hardware system, considering various factors such as scalability, reliability, performance, maintainability, and security. It serves as a blueprint for building both the software and hardware elements of a project. System design focuses on how the system will work, including how different parts interact with each other. It typically involves high-level design (HLD) for the overall system architecture and low-level design (LLD) for individual components. A good system design ensures scalability, performance, security, and maintainability. It also considers user requirements, resource constraints, and potential future changes. Effective system design is essential for building reliable, efficient, and user-friendly systems.

3.2.1. Architectural Design

We have developed the required system that makes use of the internet and computer peripherals. This system is supported by any different devices that can access the internet. The system architecture is clearly defined by the following figure:

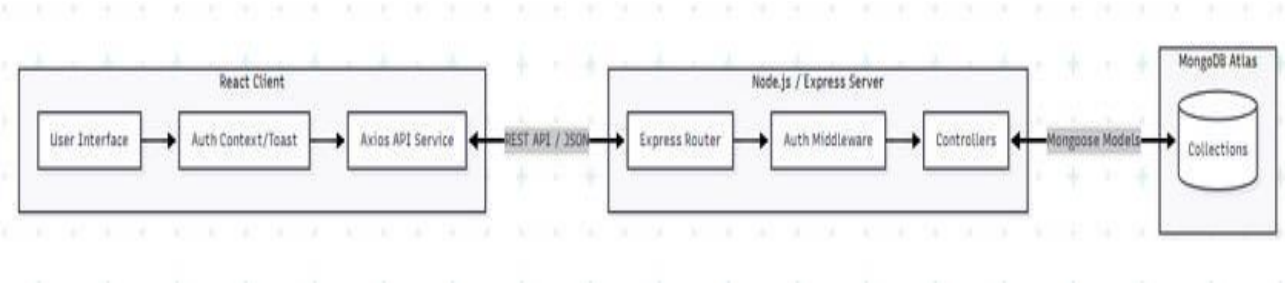


Fig 3.2.1: System Architecture

3.2.2 Database Schema Design

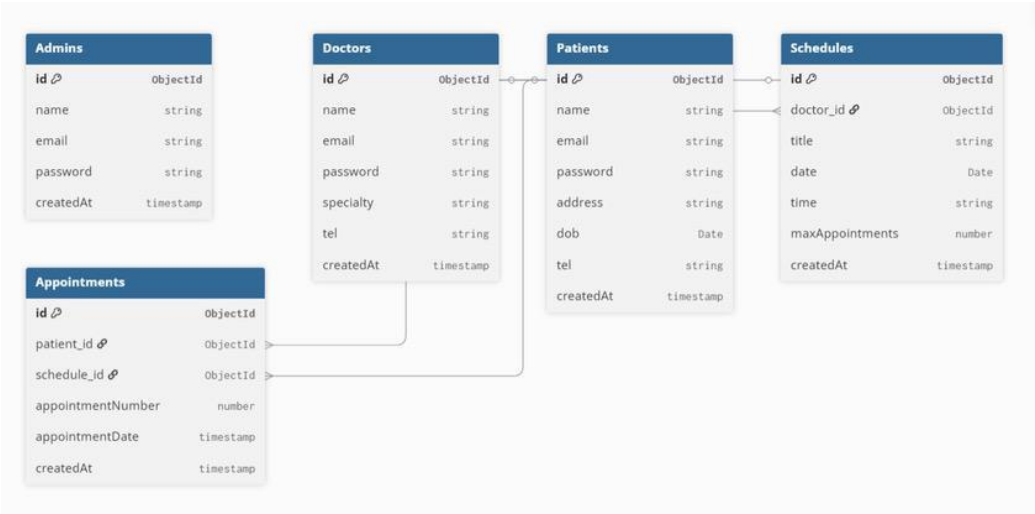


Figure 3.2.2 Database Schema Design of Online Doctor Appointment System

3.2.3 Interface Design (UI/UX)

Interface design, also known as user interface (UI) design, involves designing the visual layout of software applications or websites to ensure a smooth and intuitive user experience. This process includes creating the graphical elements, such as buttons, menus, and other interactive components, as well as defining the overall structure and navigation flow.

Interface structure diagrams are visual representations that depict the organization and flow of a user interface. They typically include elements like screens, pages, and components, along with the connections and transitions between them. These diagrams help designers and developers understand the layout and behavior of the interface, facilitating collaboration and ensuring consistency throughout the design process.

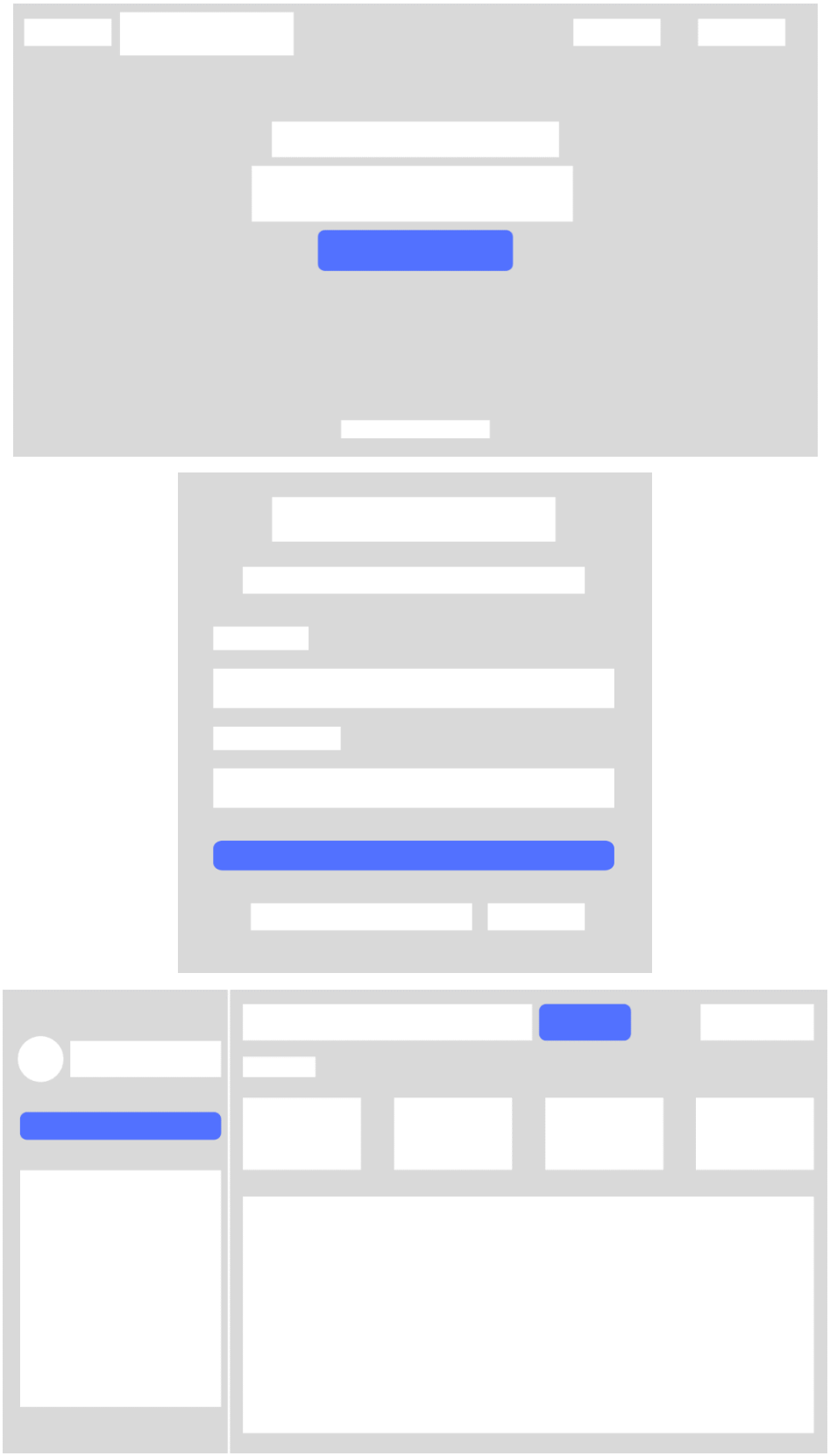


Figure:3.2.3 Interface Design

3.2.4 Physical DFD

A physical data flow diagram (DFD) of an online doctor appointment system shows how the system is actually implemented using hardware, software, and real-world data movement. It includes specific technologies like web servers, mobile apps, databases, and APIs that handle user interactions and data processing.

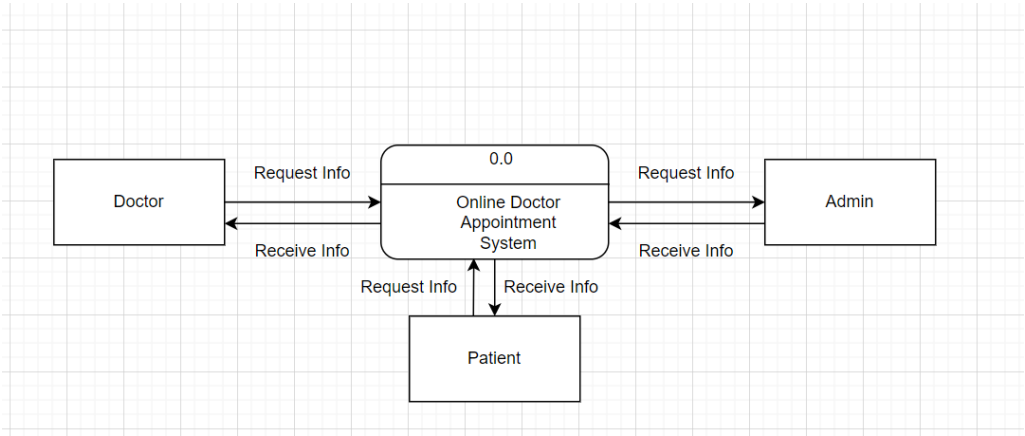


Fig 3.2.4 (a): Level 0 DFD Diagram

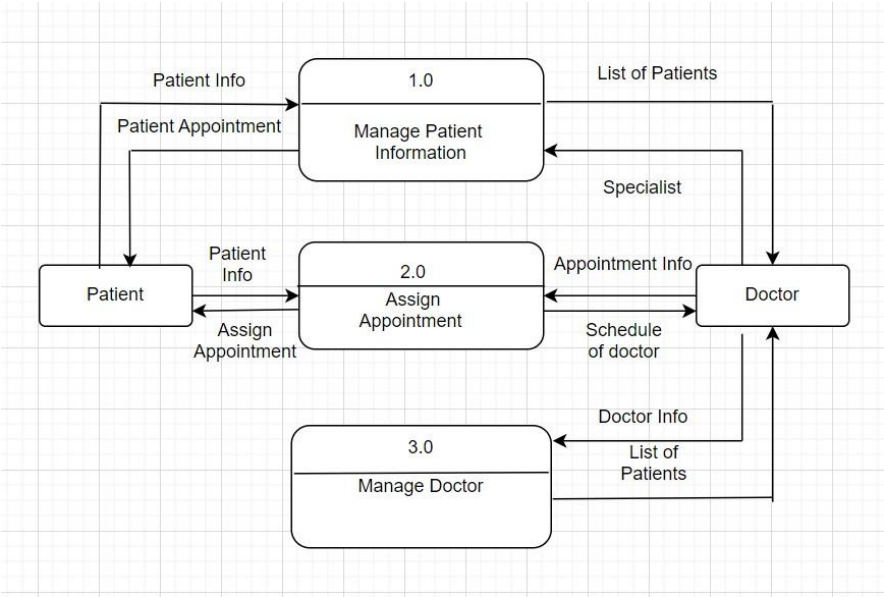


Fig 3.2.4 (b): Level 1 DFD Diagram

3.5 Description of Algorithm

Used 6 key algorithms and architectural patterns to ensure security, reliability, and efficiency:

1. First-Come, First-Served (FCFS) Scheduling

This is the primary algorithm for the booking system. It treats every appointment request in the order it is received, ensuring fairness in health service delivery.

2. Boundary Check (Capacity Management)

Used whenever a patient tries to book a session. The system runs a Counting Algorithm against the database to compare the current bookings to the maxAppointments limit. If the count matches the limit, the booking is rejected.

3. Linear Search & Filtering

Implemented in the "All Doctors" and "Scheduled Sessions" screens. It uses a case-insensitive string matching algorithm to filter large lists of data based on user input (Doc Name, Specialty, or Session Title).

4. JWT Authentication Pattern

A security algorithm that signs and verifies user sessions. It ensures that every request to the backend is authenticated using a Cryptographic Hashing Algorithm (HMAC).

5. Role-Based Access Control (RBAC)

A logic-flow algorithm that determines user permissions. It acts as a Gatekeeper Pattern, redirecting users or blocking API requests based on their privilege level (Admin vs. Doctor vs. Patient).

6. Duplicate Prevention (Set Membership)

A logical check performed in the bookAppointment controller. It searches for an existing (PatientID, SessionID) combination to ensure that the appointment set is "unique," preventing a single patient from holding more than one spot in a single session.

Chapter 4

Implementation and Testing

4.1. Implementation

Implementation is the process of implementation software through successive phases in an orderly way. This process includes not only the actual writing of code but also the preparation of requirements and objectives, the design of what is to be coded, and confirmation that what is implemented has met objectives.

4.1.1 Tools Used

Many tools were used for the development of this system. Some were used more than others, and some were used just to make the work slightly easier. These tools are described below:

CASE Tools

Diagrams.net

Diagrams.net (previously draw.io) is a web application for designing software systems. It is used in all areas of design from Model diagrams, process diagrams, flowcharts diagrams and so on of this project. PNG format was exported from the web application to be used in the project and system design.

Front End

A "front-end" application is one that application users interact with directly. The front end is an interface between the user and the back end. In simple words we can understand the concept of front and back end with the help of below given definition. In client/server applications, the client part of the program is often called the front end and the server part is called the back end. While creating a front end, different components relating to the software development were used. They are listed below:

- **PHP:**
- **HTML:**

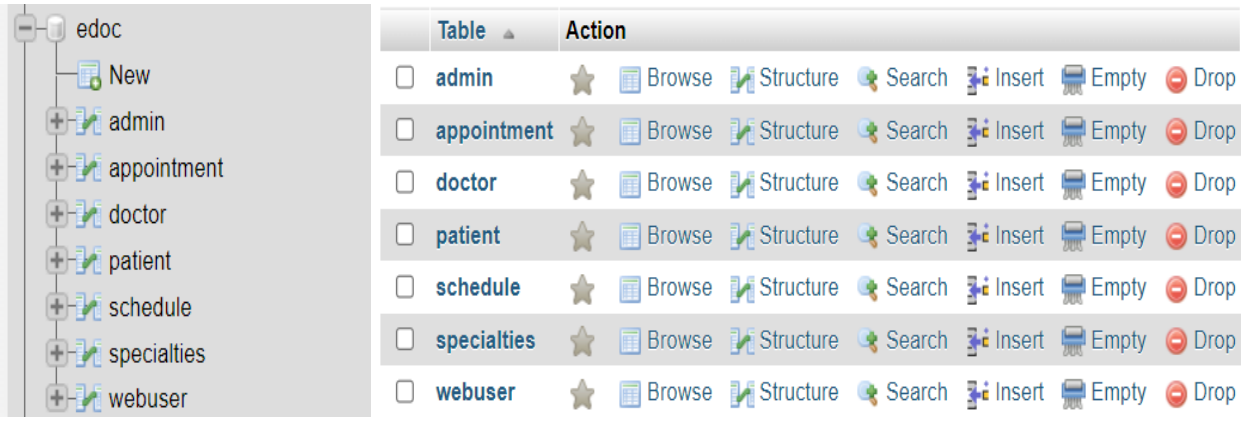
Back End

A "back-end" application or program serves indirectly in support of the front-end services, usually by being closer to the required resource or having the capability to communicate with the required resource. The back-end application may interact directly with the front-end or, perhaps more typically, is a program called from an intermediate program that mediates front-end and back-end

activities. It is the part of the system that normal users don't want to know about it. System administrator manages it. Back end possess database and administrator manages it. The following parts are contained in the back end of our system:

- **MySQL:**

MySQL is a freely available open-source Relational Database Management System (RDBMS) that uses Structured Query Language (SQL). SQL is the most popular language for adding, accessing and managing content in a database. It is most noted for its quick processing, proven reliability, ease and flexibility of use. MySQL is just the brand of one database software, one of many.



4.2. Testing

Software testing is the process of evaluation a software item to detect differences between given input and expected output. It also assesses the feature of a software item. Testing assesses the quality of the product. Software testing is a process that should be done during the development process.

We can break the definition of Software testing into the following two parts:

1. First Phase: During the software engineering that is during the module creation.
2. Second Phase: After the completion of the system/software.

4.2.1 Test Cases for Unit Testing

Each component is considered individually. It emphases on each unit of software as employed in the source code. It is white box testing. Unit testing is a software method by which individual units of source code, sets of one or more computer program modules together with associated control

data, usages producers, and operating procedures, are tested to determine whether they are valid for use. Unit testing has been done testing individual activity of the application.

Each activity like registration, sign in, reset password, dashboard, patient and doctor detail, doctor searching, book appointment, prescribe medicine for patient etc. has been tested individually. All the units testing has been done using different web browsers.

4.2.2 Test Case for System Testing:

It is executing programs to check logical deviations made in it with objective of finding errors. A system is tested for online response, capacity of transaction, recovery from failure etc. System testing is done to ensure that the system satisfies all the user requirements. System testing is the level of the software testing where a complete and integrated software is tested. The system testing has been performed by testing the whole application on a window-based computer with a local host, i.e., WAMP. The whole application run smoothly without any error.

Test cases:

- **For Patient Sign Up:**

Table 1: Test case for sign up

Test Case ID	Test Scenario	Test Steps	Input test data	Expected Results	Actual Result	Pass/ Fail
TC-1	Check signup activity with valid data	1) Open Web browser and go to patient sign up activity. 2) Fill up the form with valid data. 3) Click Signup button	a) Name: Simran Pun b) Address: Nepalgunj c) Email: simran@edoc.com d) Password: simrann e) Confirm Password: simrann	User must sign up & redirected to sign in activity	User got signed up and got login in activity	Pass

- **For Patient Sign In:**

Table 2: Test Case for Patient Sign In

Test case ID	Test Scenario	Test Steps	Test input data	Expected Results	Actual Results	Pass /Fail
TC-2	Check sign in activity with valid data	1) Open Web browser and go to patient sign in activity. 2) Fill up the form with valid data. 3) Click Sign in button	a) Username: simran@edoc.com b) Password: simrann	User should redirected to home activity	User got home activity	Pass

- **Change Password for patient:**

Table 3: Test Case for Change Password for Patient

Test Case ID	Test Scenario	Test Steps	Input test data	Expected Results	Actual Result	Pass/ Fail
TC-3	Click change password Activity	1) Open Web browser and go to home page then change password activity. 2) Input current password 3) Create new password and click on change password	a) Current password: simrann b) New password: simran c) Conform password: simran	User must get redirected to home panel	User get home panel	Pass

- **Book Appointment:**

Table 4: Test Case for Book Appointment

Test Case ID	Test Scenario	Test Steps	Input test data	Expected Results	Actual Results	Pass/fail
TC-4	Book appointment activity with valid data	1) Open Web browser and go to login activity. 2) Click book appointment and Fill up the form with valid data. 3) Click Submit button	a) Doctor Specialization: Dermatology b) Doctor Name: Bimarsh Dangi c) Consultancy fee: 2000 d) Date: 2024-06-26 e) Time: 12:00 PM	Appointment record should be submitted and user redirect to book appointment activity	As expected	Pass

- **Doctor Log In:**

Table 5: Test Case for Doctor Log In

Test case ID	Test Scenario	Test Steps	Test input data	Expected Results	Actual Results	Pass /Fail
TC-5	Check sign in activity with valid data	1) Open Web browser and go to doctor log in activity. 2) Fill up the form with valid data. 3) Click log in button	c) Username: docs@edoc.com d) Password: doct	User should redirected to home activity	User got home activity	Pass

- **Sign In for Admin**

Table 6: Test Case for Sign In for Admin

Test case ID	Test Scenario	Test Steps	Test input data	Expected Results	Actual Results	Pass /Fail
TC-7	Check sign in activity with valid data	1) Open Web browser and go to admin login activity. 2) Fill up the form with valid data. 3) Click login in button	a) Email: admin@edoc.com b) Password: admin	User should redirected to home activity	User got home activity	Pass

- **Add Doctor by Admin**

Table 7: Test Case for Add Doctor

Test Case ID	Test Scenario	Test Steps	Input test data	Expected Results	Actual Results	Pass/ fail
TC-8	Add new doctor with valid information	1) Open Web browser and go to Add Doctor activity. 2) Fill up the form with valid data. 3) Click submit button	a) Doctor Specialization: Accident and emergency medicine b) Doctor name: Doc Sims c) Consultancy fee: 2000 d) Contact number: 9816572947 e) Email: docs@edoc.com f) Password: docs	Doctor's information should be stored and redirected to add doctor activity.	As expected	Pass

Chapter 5

Conclusion and Future Recommendations

5.1 Conclusion

The proposed system will save effort, time and money of patients from waiting to book the appointment as well as reduce the work of the doctor and the hospital or clinic to access to check report and easier way to manage the appointment. The doctor and the hospital or clinic needs to maintain the record of the patients. Patients would be able to know the schedule availability on a particular date.

The implementation of the online doctor appointment system has successfully addressed the inefficiencies and challenges faced by traditional appointment scheduling methods. This system streamlines the appointment booking process, making it more convenient for both patients and healthcare providers. Patients can now book, reschedule, or cancel appointments effortlessly through a user-friendly interface, while healthcare providers can manage their schedules more efficiently, reducing no-shows and optimizing their time.

5.2 Lesson Learnt

The online doctor appointment system has taught us that accessibility to healthcare can be dramatically improved when digital tools are thoughtfully implemented. Patients no longer need to travel long distances or wait in crowded clinics, which has made medical care more inclusive and efficient. However, the success of such systems depends heavily on user-friendly design; if the interface is confusing, both patients and doctors struggle to use it effectively. Timely notifications and reminders have proven essential in reducing missed appointments and keeping schedules on track. One of the most critical lessons is the importance of data privacy—patients must feel confident that their personal health information is secure. Technical reliability is also key, as system glitches can disrupt care and erode trust. Training and support for users, especially those unfamiliar with technology, significantly improves adoption and satisfaction. Integration with electronic health records and other systems enhances continuity of care and saves time for providers. Feedback from users has shown that ongoing refinement is necessary to meet evolving

needs. Most importantly, while the system streamlines logistics, it must still preserve the human connection that defines compassionate healthcare.

5.3 Future Recommendations

To make online doctor appointment systems better in the future, they should support different languages so people from all backgrounds can use them easily. The system can use smart technology to help patients explain their health problems and find the right doctor. In places where internet is weak, it should allow booking through text messages or phone calls. Adding features like video calls with doctors, online prescriptions, and lab test bookings in one place will make things more convenient. Sending reminders through WhatsApp or SMS can help people remember their appointments. The system should also connect with hospital records so doctors can see a patient's full medical history. There should be a way to book urgent appointments quickly when someone needs help fast. It's important to make the system easy for older people and those with disabilities to use. Looking at anonymous data can help health experts understand common health problems and plan better services. Most importantly, even though it's online, the system should still feel caring and personal.

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